Introduction to Cryptography

Lecture 9: Transposition Ciphers
Rail Fence, Columnar Transposition and Double Transposition

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Transposition Cipher

Definition

A transposition cipher does not replace plaintext letters. Instead, it rearranges the positions of plaintext letters using a permutation rule.

- Plaintext symbols remain the same.
- Their positions are changed.
- The ciphertext is a permutation of the plaintext.
- Letter frequency remains unchanged.

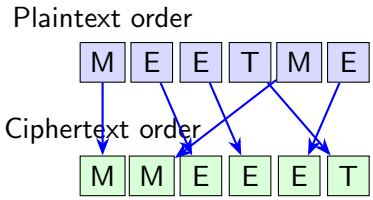
Key Idea

Substitution changes symbols; transposition changes positions.

Substitution vs Transposition

Feature	**Substitution Cipher**	**Transposition Cipher**
Main action	Replaces symbols	Rearranges symbols
Letter identity	Changes	Remains same
Letter frequency	May be preserved in simple substitution	Always preserved
Example	Caesar cipher	Rail Fence cipher
Cryptanalysis	Frequency analysis	Anagram and permutation analysis

Basic Idea of Transposition



Observation

The letters are not changed. Only their sequence is changed.

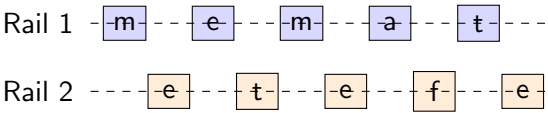
Rail Fence Cipher

Definition

Rail Fence cipher writes plaintext letters diagonally across a number of rails and then reads them row-wise to form ciphertext.

- It is the simplest transposition cipher.
- The number of rails is the key.
- For depth 2, letters are written alternately on two rows.
- Ciphertext is read row by row.

Rail Fence Cipher: Depth 2 Pattern



Rule

Write letters diagonally across rails and then read Rail 1 followed by Rail 2.

Solved Numerical 1: Rail Fence Encryption

Encrypt the plaintext:

`meet me after the toga party`

using Rail Fence cipher with depth 2.

Remove spaces:

`meetmeafterthetogaparty`

Step

Write alternate letters on Rail 1 and Rail 2.

Rail Fence Encryption Table

Rail 1:	*m*	*e*	*m*	*a*	*t*	*r*	*h*	*t*	*g*	*p*	*r*
	y										
Rail 2:	*e*	*t*	*e*	*f*	*e*	*t*	*e*	*o*	*a*	*a*	*t*

Read row-wise:

MEMATRHTGPRYETEFETEOAAT

Ciphertext

MEMATRHTGPRYETEFETEOAAT

Solved Numerical 2: Rail Fence Decryption

Decrypt the ciphertext:

MEMATRHTGPRYETEFETEOAAT

using Rail Fence depth 2.

- Total letters = 24
- Rail 1 gets 12 letters
- Rail 2 gets 12 letters

Rail 1 = MEMATRHTGPRY

Rail 2 = ETEFETEOAAT

Rail Fence Decryption Process

Interleave letters from Rail 1 and Rail 2:

Rail 1: *m e m a t r h t g p r*
y
Rail 2: *e t e f e t e o a a t*

m e e t m e a f t e r t h e t o g a p a r t y

Plaintext

meet me after the toga party

Practice: Rail Fence Cipher

Question

Encrypt the following plaintext using Rail Fence cipher with depth 2:

`cryptography`

- Remove spaces if any.
- Place alternate letters on two rails.
- Read Rail 1 followed by Rail 2.

Practice Solution: Rail Fence Cipher

Plaintext:

`cryptography`

Rail 1: $c \quad y \quad t \quad g \quad a$
h
Rail 2: $r \quad p \quad o \quad r \quad p$
y

Ciphertext

`CYTGAHRPORPY`

Columnar Transposition Cipher

Definition

In columnar transposition, plaintext is written row by row in a rectangle and ciphertext is read column by column according to a key order.

- The key defines the order of columns.
- Plaintext is filled row-wise.
- Ciphertext is generated column-wise.
- Padding may be added to complete the rectangle.

Columnar Transposition: Key Concept

Key	4	3	1	2	5	6	7
	a	*t*	*t*	*a*	*c*	*k*	*p*
	o	*s*	*t*	*p*	*o*	*n*	*e*
	d	*u*	*n*	*t*	*i*	*l*	*t*
	w	*o*	*a*	*m*	*x*	*y*	*z*

Reading Order

Read columns labelled 1, 2, 3, 4, 5, 6, 7.

Solved Numerical 3: Columnar Encryption

Encrypt:

`attack postponed until two am`

using key:

`4312567`

Remove spaces and add padding if required:

`attackpostponeduntiltwoamxyz`

Step

Write the plaintext row-wise under the key.

Columnar Encryption Matrix

Key	4	3	1	2	5	6	7
	a	*t*	*t*	*a*	*c*	*k*	*p*
	o	*s*	*t*	*p*	*o*	*n*	*e*
	d	*u*	*n*	*t*	*i*	*l*	*t*
	w	*o*	*a*	*m*	*x*	*y*	*z*

Read Columns

Column 1, then 2, then 3, then 4, then 5, 6, and 7.

Columnar Encryption Output

- Key 1 is column 3: `TTNA`
- Key 2 is column 4: `APTM`
- Key 3 is column 2: `TSUO`
- Key 4 is column 1: `AODW`
- Key 5 is column 5: `COIX`
- Key 6 is column 6: `KNLY`
- Key 7 is column 7: `PETZ`

Ciphertext

`TTNAAPTMTSUOAODWCOIXKNLYPETZ`

Columnar Transposition Algorithm

1. Choose a numeric key representing column order.
2. Remove spaces from plaintext.
3. Write plaintext row-wise in a rectangular matrix.
4. Add padding letters if necessary.
5. Read columns according to ascending key values.
6. Concatenate the columns to obtain ciphertext.

Solved Numerical 4: Columnar Decryption

Decrypt:

TTNAAPTMTSUOAODWCOIXKNLYPETZ

using key:

4312567

- Ciphertext length = 28
- Number of columns = 7
- Number of rows = 4
- Each column contains 4 letters

Columnar Decryption: Split Ciphertext

Split ciphertext into groups of 4 letters:

$$\texttt{TTNA} \quad \texttt{APTM} \quad \texttt{TSUO} \quad \texttt{AODW} \quad \texttt{COIX} \quad \texttt{KNLY} \quad \texttt{PETZ}$$

Assign them to columns in key order:

$$1 \rightarrow \texttt{TTNA}, \quad 2 \rightarrow \texttt{APTM}, \quad 3 \rightarrow \texttt{TSUO},$$

$$4 \rightarrow \texttt{AODW}, \quad 5 \rightarrow \texttt{COIX}, \quad 6 \rightarrow \texttt{KNLY}, \quad 7 \rightarrow \texttt{PETZ}$$

Columnar Decryption Matrix

Place columns back under the original key 4 3 1 2 5 6 7:

Key	4	3	1	2	5	6	7
	a	*t*	*t*	*a*	*c*	*k*	*p*
	o	*s*	*t*	*p*	*o*	*n*	*e*
	d	*u*	*n*	*t*	*i*	*l*	*t*
	w	*o*	*a*	*m*	*x*	*y*	*z*

Read row-wise:

`attackpostponeduntiltwoamxyz`

Plaintext

`attack postponed until two am`

Double Transposition Cipher

Definition

Double transposition applies the transposition process twice, usually with the same or different keys.

- First transposition rearranges the plaintext.
- Second transposition rearranges the result again.
- It creates a more complex permutation.
- It is much harder to reconstruct than single transposition.

Solved Numerical 5: Double Transposition

First transposition output:

TTNAAPTMTSUOAODWCOIXKNLYPETZ

Write again row-wise using key:

4312567

Key	4	3	1	2	5	6	7
	t	*t*	*n*	*a*	*a*	*p*	*t*
	m	*t*	*s*	*u*	*o*	*a*	*o*
	d	*w*	*c*	*o*	*i*	*x*	*k*
	n	*l*	*y*	*p*	*e*	*t*	*z*

Double Transposition Output

Read columns in key order:

- Column 1: `NSCY`
- Column 2: `AUOP`
- Column 3: `TTWL`
- Column 4: `TMDN`
- Column 5: `AOIE`
- Column 6: `PAXT`
- Column 7: `TOKZ`

Final Ciphertext

`NSCYAUOPTTWLTMDNAOIEPAXTTOKZ`

Why Double Transposition is Stronger

- Single transposition has visible structure.
- Column positions can often be guessed using digrams and trigrams.
- Double transposition creates a more complex permutation.
- The relationship between neighboring plaintext letters becomes harder to identify.

Important

Double transposition does not change letters, but it hides their original order more effectively.

Permutation View of Transposition

Plaintext letter positions:

01 02 03 04 05 06 07 08 09 10 11 12 13 14

15 16 17 18 19 20 21 22 23 24 25 26 27 28

After first transposition:

03 10 17 24 04 11 18 25 02 09 16 23 01 08

15 22 05 12 19 26 06 13 20 27 07 14 21 28

Double Transposition Permutation

After second transposition:

17 09 05 27 24 16 12 07 10 02 22 20 03 25

15 13 04 23 19 14 11 01 26 21 18 08 06 28

Observation

The second transposition produces a less regular and more difficult permutation.

Cryptanalysis of Transposition Ciphers

A pure transposition cipher is recognizable because:

- It has the same letter frequencies as the plaintext.
- Only positions are changed.
- Common letters remain common.
- Digram and trigram structure may help reconstruct the order.

Cryptanalytic Method

Try arranging ciphertext in matrices and test possible column permutations.

Letter Frequency Property

Plaintext

`attackpostponed`

Contains letters:

- t appears multiple times
- o appears multiple times
- a appears multiple times

Transposition Ciphertext

`TTNAAPTMTSUO...`

Same letters appear, but in different positions.

Key Point

Transposition preserves frequency distribution.

Comparison of Transposition Techniques

Feature	**Rail Fence**	**Columnar**	**Double Transposition**
Key	Number of rails	Column order	One/two column keys
Writing method	Diagonal	Row-wise matrix	Two-stage matrix
Reading method	Row-wise	Column-wise by key	Column-wise twice
Security	Very weak	Better	Stronger
Weakness	Easy pattern	Column guessing	Still preserves frequency

Practice Question 1

Encrypt the plaintext using Rail Fence cipher with depth 2:

`networksecurity`

Instructions

1. Write alternate letters on Rail 1 and Rail 2.
2. Read Rail 1 followed by Rail 2.
3. Write the final ciphertext.

Practice Question 1: Answer

Plaintext:

networksecurity

Rail 1: $n \quad t \quad o \quad k \quad e \quad u \quad i \quad y$

Rail 2: $e \quad w \quad r \quad s \quad c \quad r \quad t$

Ciphertext

NTOKEUIYEWRSCRT

Practice Question 2

Encrypt the plaintext:

`defend the east wall`

using columnar transposition key:

`3142`

Instructions

1. Remove spaces.
2. Write row-wise in 4 columns.
3. Add padding if needed.
4. Read columns in key order.

Practice Question 2: Solution Matrix

Plaintext after removing spaces:

```
defendtheeastwall
```

Add padding:

```
defendtheastwallx
```

Key	3	1	4	2
	d	*e*	*f*	*e*
	n	*d*	*t*	*h*
	e	*a*	*s*	*t*
	w	*a*	*l*	*l*
	x	*x*	*x*	*x*

Practice Question 2: Ciphertext

Read by key order:

- Key 1 is column 2: `EDAAX`
- Key 2 is column 4: `EHTLX`
- Key 3 is column 1: `DNEWX`
- Key 4 is column 3: `FTSLX`

Ciphertext

`EDAAXEHTLXDNEWXFTSLX`

Common Mistakes

- Forgetting to remove spaces before encryption.
- Using 1-based and 0-based positions incorrectly.
- Reading rows instead of columns in columnar transposition.
- Not adding padding to complete the matrix.
- Confusing substitution with transposition.
- Assuming transposition changes letter frequency.

Quick Quiz

1. Define transposition cipher.
2. How is transposition different from substitution?
3. What is the key in Rail Fence cipher?
4. Encrypt `attack` using Rail Fence depth 2.
5. What is the key in columnar transposition?
6. Why are pure transposition ciphers recognizable?
7. Why is double transposition stronger than single transposition?

References

- William Stallings, *Cryptography and Network Security: Principles and Practice*.
- William Stallings and Lawrie Brown, *Computer Security: Principles and Practice*.
- Douglas R. Stinson, *Cryptography: Theory and Practice*.
- Menezes, van Oorschot, and Vanstone, *Handbook of Applied Cryptography*.

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- Clarity of concepts
- Quality of examples
- Pace of the lecture
- Suggestions for improvement

Thank you for your valuable response.



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Questions and Discussion